Second Year Project

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1 Business summary

1.1 Price forecasting

Our analysis starts by forecasting gas prices, as they are a growing share of shipping costs. Being able to anticipate on this will help with budgeting and timing of shipment vessels. We tested multiple approaches to forecasting gas prices, ranging from simple trend-based models, to more complex ones that factor in consumer sentiment and inflation expectations. Let us call them model A (simple trend-based model), model B (trend-based including seasonal pattern), and model C (also including economic factors). From this we found that model A (simple historical pattern) works best compared to models B and C, as it gave us the most accurate forecast. including seasonal patterns and economic factors made the models worse. This insight is slightly counter-intuitive, as it implies more sophistication does not mean a better forecast.

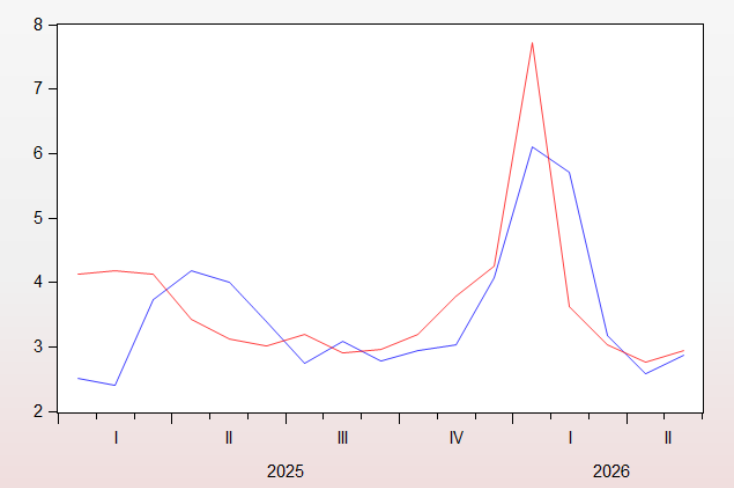


Figure 1: Forecasted price of 1 million BTU in dollars (blue) vs actual price (red), period 2025M01 - 2026M05

However, the statistical test showed the difference between the 3 models was not great enough to conclude one is far superior over the other. We thus have to conclude model A is our best estimate, not a guaranteed winner. The forecasted price of 1 million BTU in dollars (Henry Hub natural gas spot price) for the remainder of 2026 can be seen in the table below. As we can see, prices are forecast to remain relatively stable throughout the year, with a big increase towards the end.

Month	Forecast
June	2.991
July	2.884
August	2.527
September	2.905
October	2.983
November	3.037
December	3.956

Table 1: Forecasting Results

1.2 Vessel scheduling

Each month, the company delivers roughly 488.5 Kilometric tons of chemicals around twenty ports, using various types of vessel and contracts. We want to minimize the costs of this as much as possible. For our optimization of the shipment scheduling, we first built a step-by-step planning method that groups orders by region, fills ships efficiently, and assigns routes. Then it adds multiple constraints such as deadlines, port depths, and temperature requirements. In stage 1, we use this planning method without the deadlines and cargo restrictions, to find the cheapest possible way to move all cargo. we find a total cost of $12.58 million. However, without constraints, this is not very realistic. In stage 2, we add multiple constraints. each order has a delivery window (7 days early or late is acceptable, else there will be a fine), ships can't load hot and cold products next to each other, and ports have depth limits some larger or heavier ships can not enter. With this, we receive costs of about $15.78 million, around a 25% increase. This jump largely comes from unavoidable late fees, not a systematic error. These fees are incurred as the planning period starts June first, while some vessels should depart in May to avoid late fees. In stage three, we apply a fine-tuning method that looks for opportunities to move individual deliveries between different voyages if it lowers costs. We get $ 15.75 million. This save is minimal as our stage two method is already greatly optimized. The company might consider a trade-off between vessel speed of 16 instead of 15 knots, which will incur a cost of about $25k-32k in fuel, but this might be worthwhile compared to all the late fees if many ports on one voyage are at risk of being late.

Stage	Costs	Change
Stage 1	$12,58 million	/
Stage 2	$15,78 million	+25%
Stage 3	$15,75 million	-0.2%

1.3 Conclusions

We can take away from the forecast that simpler models will yield the best predictions going forward. Furthermore, we found that much of the extra costs

within our voyage scheduling comes from unavoidable late fees, which might point towards a contract timing issue rather than route inefficiencies. A logical next step would be to implement the forecasted gas prices into the planning method to find optimal vessel speeds given the predicted prices.

2 Introduction

The petrochemical industry is a highly dynamic environment, where companies have to make impactful decisions under uncertainty, all while trying to maintain operations at a cost-effective level. Two of these challenges include accurately forecasting energy prices, and scheduling the transportation of chemicals across the globe. A reliable forecast can support financial and operational decisions, while efficiently scheduling will result in minimized transport costs and on-time deliveries.

Our report will combine multiple techniques in econometrics and operations research to achieve just this. In the first section, we will perform an analysis on the dynamics on the prices of US natural gas, and worldwide crude oil. We will transform the time series appropriately before estimating a number of models (AR, ARMA, VAR), before deciding on the best to use in a forecast for the future prices of both series.

The second part will investigate and optimize the scheduling of vessels for an international petrochemical company. We will first formulate an algorithm designed to find the optimal shipping routes, before extending it to include fuel costs, delivery dates, vessel capacity & cargo compatibility.

Our report aims to illustrate how econometrics and operations research can be applied in real-life scenarios by applying both to two of the biggest issues within a petrochemical shipping company.

3 An investigation of oil and gas prices

3.1 ARMA Forecast

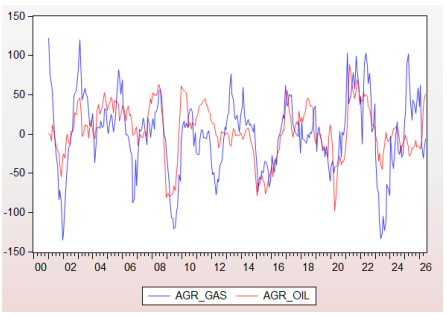
Our goal in this section will be to find a model best fit to forecast the prices of the 2 series we chose. To do this, we will make use of, firstly, AR(p) models, where we will select the most accurate one per series using the BIC. Then, we will test several ARMA and SARMA models to see if they produce better forecasts. Before be begin our ARMA estimations, we must first make sure our data is stationary, as ARMA assumes that the probability distribution of the price data does not change as time moves forward. We accomplish this with two transformations per the two time series.

The first transformation is the annual growth rate per month.

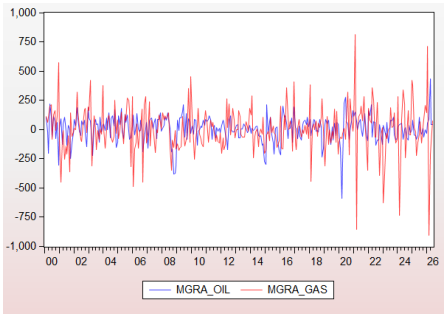
$$\pi_{a,t} = 100 \times \ln\left(\frac{P_t}{P_{t-12}}\right) \tag{1}$$

The second is the annualized monthly growth rate.

$$\pi_{m,t} = 1200 \times \ln\left(\frac{P_t}{P_{t-1}}\right) \tag{2}$$



(a) Annual growth rate



(b) Monthly growth rate annualized

Figure 2: Graphs of both transformations, 2000M01 - 2026M05

To confirm all series are stationary, we perform an ADF test. The results we can see in the table below. For all four series, the test-statistic is smaller than the 1% critical value, meaning we reject the null of the series having a unit root. Thus, we can conclude all 4 series are stationary.

Series	T-stat	1% Critical Value	Conclusion
agr_gas	-3.73	-3.45	Stationary
agr_oil	-4.62	-3.45	Stationary
mgra_gas	-17.60	-3.45	Stationary
mgra_oil	-12.94	-3.45	Stationary

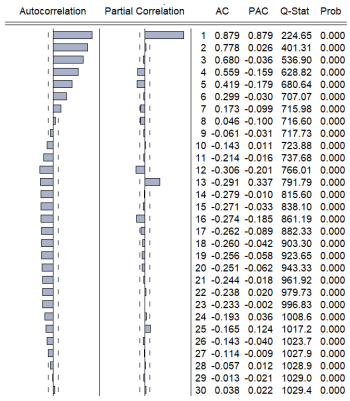
Table 2: ADF results for 4 series

3.1.1 AR(p) model estimation

An AR(p) model says todays growth rates depend on the rates of the last t months:

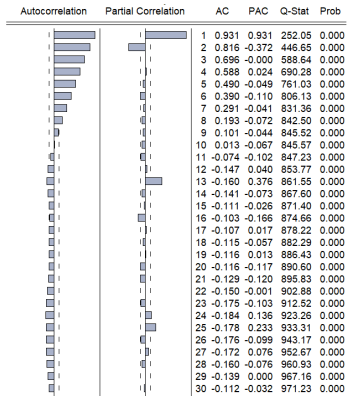
$$\pi_t = \alpha_0 + \alpha_1 \pi_{t-1} + \cdots + \alpha_p \pi_{t-p} + \varepsilon_t \tag{3}$$

As requested, we estimate the p of the model using the information criteria. In the interest of parsimony, we select our best AR(p) model using the BIC, since it penalizes unnecessary regressors the most. We check the BIC of every series up until p=25, and we find the results below as optimal per series. These are the models we are gonna use in the forecast.



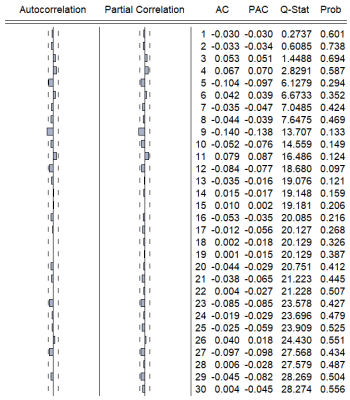
	AC	PAC	Q-Stat	Prob
1	0.879	0.879	224.65	0.000
2	0.778	0.026	401.31	0.000
3	0.680	-0.036	536.90	0.000
4	0.559	-0.159	628.82	0.000
5	0.419	-0.179	680.64	0.000
6	0.299	-0.030	707.07	0.000
7	0.173	-0.099	715.98	0.000
8	0.046	-0.100	716.60	0.000
9	-0.061	-0.031	717.73	0.000
10	-0.143	0.011	723.88	0.000
11	-0.214	-0.016	737.68	0.000
12	-0.306	-0.201	766.01	0.000
13	-0.291	0.337	791.79	0.000
14	-0.279	-0.010	815.60	0.000
15	-0.271	-0.033	838.10	0.000
16	-0.274	-0.185	861.19	0.000
17	-0.262	-0.089	882.33	0.000
18	-0.260	-0.042	903.30	0.000
19	-0.256	-0.058	923.65	0.000
20	-0.251	-0.062	943.33	0.000
21	-0.244	-0.018	961.92	0.000
22	-0.238	0.020	979.73	0.000
23	-0.233	-0.002	996.83	0.000
24	-0.193	0.036	1008.6	0.000
25	-0.165	0.124	1017.2	0.000
26	-0.143	-0.040	1023.7	0.000
27	-0.114	-0.009	1027.9	0.000
28	-0.057	0.012	1028.9	0.000
29	-0.013	-0.021	1029.0	0.000
30	0.038	0.022	1029.4	0.000

AGR_GAS



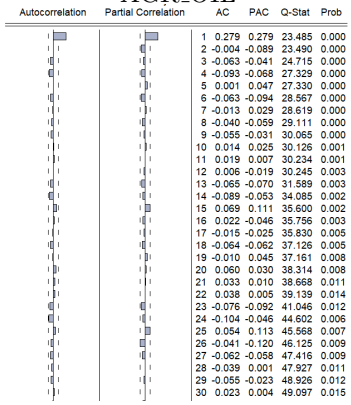
	AC	PAC	Q-Stat	Prob
1	0.931	0.931	252.05	0.000
2	0.816	-0.372	446.65	0.000
3	0.696	-0.000	588.64	0.000
4	0.588	0.024	690.28	0.000
5	0.490	-0.049	761.03	0.000
6	0.390	-0.110	806.13	0.000
7	0.291	-0.041	831.36	0.000
8	0.193	-0.072	842.50	0.000
9	0.101	-0.044	845.52	0.000
10	0.013	-0.067	845.57	0.000
11	-0.074	-0.102	847.23	0.000
12	-0.147	0.040	853.77	0.000
13	-0.160	0.376	861.55	0.000
14	-0.141	-0.073	867.60	0.000
15	-0.111	-0.026	871.40	0.000
16	-0.103	-0.166	874.66	0.000
17	-0.107	0.017	878.22	0.000
18	-0.115	-0.057	882.29	0.000
19	-0.116	0.013	886.43	0.000
20	-0.116	-0.117	890.60	0.000
21	-0.129	-0.120	895.83	0.000
22	-0.150	-0.001	902.88	0.000
23	-0.175	-0.103	912.52	0.000
24	-0.184	0.136	923.26	0.000
25	-0.178	0.233	933.31	0.000
26	-0.176	-0.099	943.17	0.000
27	-0.172	0.076	952.67	0.000
28	-0.160	-0.076	960.93	0.000
29	-0.139	0.000	967.16	0.000
30	-0.112	-0.032	971.23	0.000

AGR_OIL



	AC	PAC	Q-Stat	Prob
1	-0.030	-0.030	0.2737	0.601
2	-0.033	-0.034	0.6085	0.738
3	0.053	0.051	1.4488	0.694
4	0.067	0.070	2.8291	0.587
5	-0.104	-0.097	6.1279	0.294
6	0.042	0.039	6.6733	0.352
7	-0.035	-0.047	7.0485	0.424
8	-0.044	-0.039	7.6475	0.469
9	-0.140	-0.138	13.707	0.133
10	-0.052	-0.076	14.559	0.149
11	0.079	0.087	16.486	0.124
12	-0.084	-0.077	18.680	0.097
13	-0.035	-0.016	19.076	0.121
14	0.015	-0.017	19.148	0.159
15	0.010	0.002	19.181	0.206
16	-0.053	-0.035	20.085	0.216
17	-0.012	-0.056	20.127	0.268
18	0.002	-0.018	20.129	0.326
19	0.001	-0.015	20.129	0.387
20	-0.044	-0.029	20.751	0.412
21	-0.038	-0.065	21.223	0.445
22	0.004	-0.027	21.228	0.507
23	-0.085	-0.085	23.578	0.427
24	-0.019	-0.029	23.696	0.479
25	-0.025	-0.059	23.909	0.525
26	0.040	0.018	24.430	0.551
27	-0.097	-0.098	27.568	0.434
28	0.006	-0.028	27.579	0.487
29	-0.045	-0.082	28.269	0.504
30	0.004	-0.045	28.274	0.556

MGRA_GAS



	AC	PAC	Q-Stat	Prob
1	0.279	0.279	23.485	0.000
2	-0.004	-0.089	23.490	0.000
3	-0.063	-0.041	24.715	0.000
4	-0.093	-0.068	27.329	0.000
5	0.001	0.047	27.330	0.000
6	-0.063	-0.094	28.567	0.000
7	-0.013	0.029	28.619	0.000
8	-0.040	-0.059	29.111	0.000
9	-0.055	-0.031	30.065	0.000
10	0.014	0.025	30.126	0.001
11	0.019	0.007	30.234	0.001
12	0.006	-0.019	30.245	0.003
13	-0.065	-0.070	31.589	0.003
14	-0.089	-0.053	34.085	0.002
15	0.069	0.111	35.600	0.002
16	0.022	-0.046	35.756	0.003
17	-0.015	-0.025	35.830	0.005
18	-0.064	-0.062	37.126	0.005
19	-0.010	0.045	37.161	0.008
20	0.060	0.030	38.314	0.008
21	0.033	0.010	38.668	0.011
22	0.038	0.005	39.139	0.014
23	-0.076	-0.092	41.046	0.012
24	-0.104	-0.046	44.602	0.006
25	0.054	0.113	45.568	0.007
26	-0.041	-0.120	46.125	0.009
27	-0.062	-0.058	47.416	0.009
28	-0.039	0.001	47.927	0.011
29	-0.055	-0.023	48.926	0.012
30	0.023	0.004	49.097	0.015

MGRA_OIL

Figure 3: ACF and PACF, 30 lags

Series	Lowest BIC	Optimal Lag
agr_gas	8.886	13
agr_oil	7.836	13
mgra_gas	13.306	0
mgra_oil	12.190	1

Table 3: Optimal lag selection based on minimum BIC

3.1.2 ARMA(p,q) model estimation

To find our best ARMA model, we require stationarity, which we checked before. Next, we use the ACF and PACF at each time series to identify which p and q are optimal, and to find any eventual seasonality we should add. All PACF and ACF values for the 4 series can be found in figure 2.

AGR_Gas. We see a slowly decaying ACF, and a cutoff at lag 1 for the PACF. this all points to AR(1). We can also see a tiny seasonal component with the PACF spikes at 13 and 25, suggesting a seasonality at lag 12. This suggests, according to theory, an inclusion of SAR(1)12 will be correct. When we estimate the second model, we find another ACF/PACF spike at 24, suggesting SAR(2)12 might help. We thus will test 3 models:

Model	Significant?	BIC	Residual Diagnostics
AR(1)	Yes ($p = 0$)	9.073	Spike at lag 13 & high q-stats from lag 12 onward
AR(1) $\times$ SAR(1)$_{12}$	Yes ($p = 0$)	8.790	No major spikes in first 36 lags except lag 24
AR(1) $\times$ SAR(2)$_{12}$	Yes ($p = 0$)	**8.720**	Small q-values all along

Table 4: AGR_Gas model comparison

We select the AR(1) $\times$ SAR(2)$_{12}$ model based on the lowest BIC and the improved residual diagnostics (low q-statistics)

AGR_Oil. The ACF shows slow decay like agr_gas, while the PACF shows a significant spike at lags 1 and a smaller spike at 2, suggesting either an AR(1) or AR(2). Seasonal effects are again present at lag 12 & 25, motivating us to again include SAR(1)12 and eventually SAR(2)12. We test AR(1) and AR(2), with and without SAR(1)12. The best one among the AR(p) models we run with SAR(2)12 as well.

Model	Significant?	BIC	Residual Diagnostics
AR(1)	Yes ($p \approx 0$)	7.980	Big Seasonal spikes at lag 12
AR(2)	Yes ($p \approx 0$)	7.851	Low q-stats except seasonal spikes at 12&24
AR(1) $\times$ SAR(1)$_{12}$	Yes ($p = 0$)	7.798	Remaining q-stats issues
AR(2) $\times$ SAR(1)$_{12}$	Yes ($p = 0$)	7.701	Low q-stats, but spike at lag 24
AR(2) $\times$ SAR(2)$_{12}$	Yes ($p = 0$)	**7.587**	No major spikes; well-behaved residuals

Table 5: AGR_OIL model comparison

We select the AR(2) $\times$ SAR(2)$_{12}$ model, stating the same reasons as agr_gas. For both AGR series, a small almost insignificant spike remains at lag 36. We estimated for both a model included SAR(3)12, but the diagnostics and information criteria show that, although the model improves by a tiny bit in BIC, and remains the same in q-stats, we do not include SAR(3)12 in our model. We do this to prioritise parsimony and overall residual performance, and accept this as a known limitation.

MGRA_Gas. All spikes are not, or only barely significant. This strongly suggests white noise. For completeness, we will, in addition to ARMA(0,0) (constant), check AR(1) and MA(1).

Model	Significant?	BIC	Residual Diagnostics
Constant	–	**13.306**	No significant spikes, low q-stats
AR(1)	Insignificant	13.343	No significant spikes, low q-stats
MA(1)	Insignificant	13.344	No significant spikes, low q-stats

Table 6: MGRA_GAS model comparison

We select the constant (white noise) model.

MGRA_Oil. Similar to mgra_gas. However, there are 2 significant spikes at lag 1 for both ACF and PACF. No significant seasonality either. Since they both cut off, they suggest MA(1) & AR(1). We will check these 3 models + a constant model for completeness.

Model	Significant?	BIC	Residual Diagnostics
Constant	–	12.251	Significant lag-1 spike; Q-stats persist
AR(1)	Yes ($p = 0$)	12.208	Well-behaved residuals
MA(1)	Yes ($p = 0$)	**12.202**	Best residual performance
ARMA(1,1)	Both terms become insignificant	12.221	No major issues

Table 7: MGRA_OIL model comparison

The AR(1) and MA(1) both look very similar in every way. However, since the BIC is a bit smaller at MA(1), we will take MA(1).

3.1.3 Forecasting using selected models

We estimate all selected models from 2000M01 to 2024M12, and estimate, using static forecasting, 2025M01 to 2026M05. We then use the data from 2025M01 to 2026M05 to check how accurate our forecasted values are. Specifically, we forecast all 4 series for that interval, and then transform the forecasted values back to prices using the inverse of the transformation we used to construct the series:

$$P_t = P_{t-1} \cdot \exp\left(\frac{\pi_{m,t}}{1200}\right) \tag{4}$$

$$P_t = P_{t-12} \cdot \exp\left(\frac{\pi_{a,t}}{100}\right) \tag{5}$$

We then compare the forecasted 4 price series with the actual series using the RMSE:

$$\text{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(y_t - \hat{y}_t)^2} \tag{6}$$

In the end we are left with 8 Root mean squared errors, reported below. These values indicate how close our models are to the actual values on average. This statistic will be relevant later.

Series	AR(p)	RMSE (AR)	ARMA model	RMSE (ARMA)
mgra_gas	$p = 0$	1.3686	ARMA(0,0)	1.3686
mgra_oil	$p = 1$	7.9363	MA(1)	7.9334
agr_gas	$p = 13$	0.95336	AR(1) $\times$ SAR(2)$_{12}$	1.0876
agr_oil	$p = 13$	8.21996	AR(2) $\times$ SAR(2)$_{12}$	8.7040

Table 8: RMSE comparison between AR and ARMA/SARMA specifications

Interestingly, for both AGR series, the SARMA series yield a higher RMSE than the AR(13) models. We can explain this with the fact that AR(13) already includes much of the 12-month seasonal structure. Differently stated, the added seasonal component SAR(2)12 provides only little new info beyond what AR(13) already picks up, while introducing additional estimation uncertainty that slightly worsens the models performance outside the training sample. Nevertheless, the SARMA models are still preferred on the basis of parsimony and in-sample fit, as reflected by their lower BIC values and cleaner residual diagnostics.

3.2 Vector Autoregression (VAR)

A VAR(p) models each variable as a linear function of p lags of all variables in the system.

The difference between our VAR and the univariate ARMA and AR models is that the ARMA and AR models ignore exogenous macroeconomics expectations like consumer sentiment index and inflation expectations. A VAR model captures those variables.

We obtain the macroeconomics expectation data from FRED: University of Michigan.

Do note, when the analysis was done, we only had available data till April and the data till May was released shortly after.

3.2.1 Stationarity

After obtaining the required data, since inflation expectations and the consumer sentiment index changes over time, we first test stationarity using the Augmented Dickey Fuller (ADF) unit root test. The null hypothesis (H_0) of the ADF unit root test is that the data is non stationary.

Macroeconomic Expectation	p-value	Stationarity	Rectifying Action
Inflation Expectation	0.0010	Stationary	No Action
Consumer Sentiment Index	0.0765	Non Stationary	Took First Difference

Table 9: ADF Unit Root Test

Further on, using the ADF test again we checked whether we need to take the second difference if our remedy of taking the first difference is not station-

ary. However, with a p-value for 0.0000 for the first difference check we safely conclude that the first difference is stationary and further carry our tests.

3.2.2 Lag Length Criteria

Initially we started with a VAR(1) which includes 1 lag of each annual growth rate of gas, consumer sentiment and inflation expectation.

$$\begin{pmatrix} \pi_{gas,t} \\ sent_t \\ \pi_t^e \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} + \begin{pmatrix} \theta_{11} & \theta_{12} & \theta_{13} \\ \theta_{21} & \theta_{22} & \theta_{23} \\ \theta_{31} & \theta_{32} & \theta_{33} \end{pmatrix} \begin{pmatrix} \pi_{gas,t-1} \\ sent_{t-1} \\ \pi_{t-1}^e \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \\ \varepsilon_{3t} \end{pmatrix}$$

The VAR(1) model showed very low R^2 results across all three equations, and no coefficient was statistically significant, suggesting that a single lag is not sufficient enough to capture the dynamics of the model.

After observing the VAR 1 results, we tried to determine the optimal lag order, which was computed via View -> lag structure -> lag length criteria. The results are in the left figure below.

Lag	AIC	BIC/SC
0	18.61112	18.64890
1	15.49868	15.64978*
2	15.46887	15.73329
3	15.48144	15.85919
4	15.45481	15.94589
5	15.42759	16.03199
6	15.47663	16.19435
7	15.46314	16.29419
8	15.49113	16.43550
9	15.46929	16.52698
10	15.40815	16.57917
11	15.44229	16.72663
12	15.38977*	03.78744

Table 10: Lag Length Criteria Selection

From the above table, the AIC and BIC disagree on the amount of lags to be chosen. Given that BIC penalizes model complexity more, it is removing a lot of macroeconomic dynamics from the system. While on the other hand, AIC was choosing lag 10, would could potentially cause overfittng and make our model worse. Due to these reasons, we decide to choose a number of lags in between BIC and AIC, and thus continued our analysis with lag 2. Lag 2 balances short run dynamics while avoiding overfitting. Upon further estimation, VAR(2) gave the following results.

Dependent Variable	**VAR(1)**	**VAR(2)**
Consumer Sentiment	0.0168	0.079
Gas Return	0.002	0.785
Inflation Expectations	0.788	0.796

Table 11: Comparison of R^2 values for the VAR(1) and VAR(2) models

Comparing the VAR(1) and VAR(2) results we observe an overall higher R^2.

3.2.3 Granger Causality

Granger Causality test whether the past values of one variable X can help to predict the value of another variable Y. The null hypothesis (H_0) is that X does not Granger causes Y. Since we chose VAR(2) from the previous section, we will further only look into the VAR(2) results.

Dependent Variable	Excluded	p-value	Conclusion
Annual Gas Return	Inflation Expectation	0.7702	Not Significant
Annual Gas Return	Stationary Consumer Sentiment	0.3292	Not Significant
Inflation Expectation	Annual Gas Return	0.0364	Significant
Inflation Expectation	Stationary Consumer Sentiment	0.3944	Not Significant
Stationary Consumer Sentiment	Annual Gas Return	0.0121	Significant
Stationary Consumer Sentiment	Inflation Expectation	0.3581	Not Significant

Table 12: Granger Causality Results VAR(2)

From the above results, we observe that for inflation expectation and for the stationary consumer sentiment, only the inflation expectation is statistically significant.

The Granger causality results show that annual gas returns contain predictive information for inflation expectations, but inflation expectations do not have any predictive power over gas returns. This implies that incorporating gas returns into the forecasting model, as done in the VAR, can possibly improve forecast accuracy over univariate models.

3.2.4 VAR Forecasting and RMSE

Testing the accuracy of our model, we start by forecasting gas prices for 2025 month 01 to 2026 month 05. All models were estimated on the sample up to December 2024, and 17 one step ahead forecasts were generated for the period January 2025 to May 2026.

The forecasted results found were as follows:

Year	Month	Forecast
2025	M01	3.642
2025	M02	2.142
2025	M03	3.317
2025	M04	4.055
2025	M05	4.205
2025	M06	3.636
2025	M07	2.409
2025	M08	2.893
2025	M09	3.212
2025	M10	2.841
2025	M11	2.99
2025	M12	5.107
2026	M01	5.612
2026	M02	7.161
2026	M03	3.598
2026	M04	2.624
2026	M05	2.613

Table 13: VAR Forecasting Results

Forecast accuracy was evaluated using the Root Mean Squared Error (RMSE). The RMSE uses a similar formula as before, the only change being that T=17.

$$\text{RMSE} = \sqrt{\frac{1}{17}\sum_{t=1}^{T}(y_t - \hat{y}_t)^2} \tag{7}$$

The found RMSE's are given below:

Model	RMSE
AR	0.953
ARMA	1.088
VAR	1.251

Table 14: RMSE

The AR(13) achieved the lowest RMSE of 0.953, followed by the ARMA model with 1.088, while the VAR model performed considerably worse with an RMSE of 1.463.

3.3 Diebold Mariano and Forecasting

To formally assess whether differences in forecasting accuracy across models are statistically significant, we implement the Diebold-Mariano (DM) test for each pair of models. Following the case methodology, we regress

$$d_t = \left(\hat{\epsilon}^A_{t+1}\right)^2 - \left(\hat{\epsilon}^B_{t+1}\right)^2 \tag{8}$$

on a constant and test whether the intercept is zero. The observed results show that none of the three pairwise comparisons yield a statistically significant intercept

Models Tested	p-value
AR vs. ARMA	0.65
AR vs. VAR	0.32
ARMA vs. VAR	0.43

Table 15: Diebold Mariano p-values

Therefore, while AR(13) produces the lowest RMSE, we cannot conclude that it significantly outperforms the other models over the 17 month forecasting.

The forecasting results until the end of the year we got are shown in the figure below:

Month	Forecast
2026M06	2.990930
2026M07	2.884467
2026M08	2.526795
2026M09	2.904792
2026M10	2.983000
2026M11	3.037357
2026M12	3.955577

Table 16: Forecasting Results

The AR(13) model forecasts annual gas returns to remain stable from June to October 2026, before rising more sharply toward year end, reaching 3.96 by December 2026. As a univariate model, this forecast relies only on past gas returns and does not incorporate predictors such as inflation expectations, which the Granger causality results suggested may carry additional predictive value.

4 Vessel scheduling optimization

In the current world, where transportation of liquid chemicals is of utmost priority, it is vessel scheduling problems of vessels like time-chartered (TC) and spot vessels which are relevant to us. The choice of vessels here also represent a security of supply and flexibility.

The unit of measurement used to measure liquid chemicals is KMT which stands for "kilo metric tonne", i.e. 1000 metric tons. It is used to measure vessel capacity. This is relevant since the problem at hand is a capacity constrained routing problem and we need to meet demand and various ports. KMT is the key capacity unit used to find a viable solution.

Given the capacity constraint, the scheduling problem is about developing an algorithm that can find a solution to plan the transportation of 488.5 KMT of liquid chemicals to different ports around the world from the port of Jubail in Saudi Arabia while minimising costs. We do this by using a mix of time chartered (TC) and spot vessels. The idea of the problem is how to pack everything under certain constraints, how to assign vessels and when to depart.

Further on, we also implement local exchange to potentially improve on our found solution.

4.1 Unconstrained Shipping Optimization

Step 1: Regional Grouping

Before applying any algorithms, the optimization model must first overcome a bottleneck: the exponential growth of possible voyage sequences as the number of destination ports increases. In an unconstrained setting, the algorithm is technically free to put together any subset of the 20 destination ports together. This freedom produces economically absurd routes such as routing a vessel from Jubail to Singapore, then to Rotterdam, and back to Mumbai. Such a route would waste a lot of fuel and violate basic shipping feasibility. To exclude such an absurd result, the destination network is seperated into three geographic regions before any vessel assignment occurs.

1. Middle East/India: Jebel Ali, Karachi, Mumbai, Cochin, Paradeep, Adabiyah
2. Southeast Asia: Singapore, Penang, Bangkok, Zhuhai, Jiangyin, Ningbo, Ulsan, Hirohata
3. Europe: Antwerp, Rotterdam, Barcelona, Ravenna, Gebze, Izmir

This ensures that voyages stay geographically sensible and do not combine ports from opposite sides of the world, like Singapore and Rotterdam are not taken in one voyage.

Step 2: Aggregate Port Demand

All demands are further aggregated by destination port and product, giving a total KMT per port. For example, Antwerp needs 59 KMT total (Amines 28.5, Butanol 12, MMA 12, Styrene 6.5). This gives us a more manageable demand per stop rather than dealing with individual orders. By doing so, we are able to simplify the packing problem.

This is then summarised into a per port overview containing the total KMT, COOL KMT, and HEAT KMT, which the algorithm uses during the packing process. These steps aid us in solving the unconstrained and constrained optimization problems.

Before adding all the restrictions of the real world, we first start with solving a very simple basic case with no restrictions. In this case, we are ignoring delivery dates, temperature constraints, and port draft limits. The only goal is to minimize total cost under no constraints.

Step 3: Greedy Bin Packing

Greed Bin Packing sorts all ports per region by total demand in descending order before packing, and then it starts filling open voyages by adding ports from the same region until vessel capacity is exceeded, then it opens a new voyage. The greedy bin packing however does not consider temperature zoning, and can put hot KMT and cold next to each other.

Step 4: Nearest Neighbour Port Sequencing

After starting from Jubail, the algorithm applies greedy nearest neighbour and the voyage sails to the next closest unvisited port before ending at the final port. By going to the closest nearest neighbour, we are able to minimise total sailed distance and thus minimise cost.

Step 5: Vessel Assignment

The only condition for vessel assignment is that we should utilise all the time chartered vessels first before utilising spot vessels since we pay the hire price regardless for the TC vessels. More specifically, we utilise the MR50 TC vessels when KMT is more than 20, while for anything below 20 KMT, we utilise the J19 TC vessels. Using just the TC vessels, we are able to serve 310 KMT, and for the remaining demand of about 178.5 KMT, we can book spot vessels.

Step 6: Cost Calculation

After the algorithm has done all the previous steps, it now computes the costs and later checks for any improvements which can be made. To compute the costs, we look at the following costs incurred.

$$\text{Sailing days} = \text{total distance in nm}/(15kn * 24hr) \tag{9}$$

$$\text{Hire Cost} = \text{days} * \text{daily hire rate} \tag{10}$$

$$\text{fuel} = \text{loaded consumption} * \text{days} * 700.50/mt \tag{11}$$

$$\text{Total Cost} = \text{Hire} + \text{Fuel} \tag{12}$$

For time chartered vessels, we also need to add the return leg at empty consumption rate but that uses a lower per day consumption rate.

Step 7: Improvement Pass

The algorithm checks if any voyage is below 70% utilisation, and whether certain voyages can be merged with another voyage going in the same direction. The hire cost savings typically outweigh any extra distance sailed.

4.2 Constrained Shipping Optimization

4.2.1 Constraints

From the previous section the constraints we now add are:

1. **Laycan Window:** For an order with required delivery date D, the only acceptable arrival window is [D - 7, D + 7]. Both early and late deliveries beyond the window are penalised at $5,000/day. Arrival within the window costs nothing extra.

2. **Departure window per order:** Given a direct sailing time of S days from Jubail to port P at 15 knots, the vessel must depart Jubail between (D - 7 - S) and (D + 7 - S) for the order to arrive within its laycan. This is the feasible departure window for that order.

3. **Inevitable lateness:** The planning horizon starts on June 1, 2024, but 18 orders have latest departure dates that are already in May. These are orders for Antwerp, Rotterdam, Jiangyin, Ningbo, and Ulsan with early June delivery dates. No vessel departing on or after June 1 can deliver them on time. These 18 orders will inevitably incur late penalties regardless of the plan, and the algorithm's job is to minimise how late they are, not to pretend lateness can be avoided. This point is further discussed with the speed decision.

4. **Draft Limits:** Vessels sit deeper in the water the more cargo they carry. A draft limit ensures a vessel never enters a port too shallow for its current load, and as cargo is unloaded along a voyage, ports that were initially inaccessible can become reachable.

5. **Hold Restriction:** Our algorithm also respects hold constraints, ensuring hot and cold chemicals are not placed next to each other but at least a single neutral compartment is present in between them.

4.2.2 Constrained Shipping Plan Algorithm

Step 1: Compute Departure Windows

Since we now have an acceptable arrival window, our first step of solving this optimization problem is to calculate the direct sailing time, the open and close departure window, and the slack (window close - June 1). A negative slack means the delivery is already infeasible in the alloted time, however the algorithm still schedules them as early as possible to minimise total lateness.

Step 2: Group into departure cohorts:

All orders on the same voyage must share a departure date within all of their windows. If windows don't overlap, order split into different voyages. The chosen departure must lie inside the alloted window of every order on that voyage.

Step 3: Multi port departure dates:

When combining multiple ports on one voyage, the departure dates must for all ports, accounting for the ships arriving at each port at a different time.

Step 4: Speed Decision A solution to being inevitably late is to increase the vessel speed. By default, the speed is set to 15 knots, increasing speed from 15 to 16 knots reduces voyage time by roughly 1.2 days, but increases fuel costs by approximately $30,000 per voyage. At the same time, arriving outside the agreed laycan results in a penalty of $5,000 per day. Whether higher speed is justified, depends on a direct comparison between the additional fuel cost and the reduction in delay penalties. The decision rule therefore explicitly compares

total fuel cost against the penalty savings and select the option that minimises total cost.

Step 5: Voyage Packing (4 checks):

When packing everything into one voyage, the algorithm does the following 4 checks:

1. Cargo fits within vessel capacity
2. Cool+Heat within hold zone limits
3. Valid port ordering satisfying depth limits
4. Valid departure date gets all cargo to all ports within delivery windows, or, if impossible, splitting is cheaper.

Step 6: Two Trip Scheduling

TC ships that return to Jubail early enough can be redeployed on remaining demand, after re applying all 4 checks from step 5.

Step 7: Penalty Minimizing Departure Date

Unavoidable late voyages, find the single departure date that minimzes total days outside departure window across all orders, solved by testing "breakpoint" dates where individual orders cross the penalty threshold.

Step 8: Post Sequencing Priority

There are 3 conditions added into the algorithm, to decide which vessel goes first:

1. Draft constraints are hard, they cannot be violated.
2. Departure window timing, minimize penalty, serve the tightest deadline port as early in the sequenze as allowed by the draft constraint.
3. Distance efficieny is the tiebreaker when both draft and departure window are equally satisfied by multiple sequences.

Step 9: Hold Plan

Cool and Heat cargo stay in their respective column zones with a neutral buffer in between them.

Step 10: Cost Formula

$$\text{Total Cost} = \text{Hire Cost} + \text{Fuel Cost} + \text{days outside laycan} * \$5000/\text{order} \quad (13)$$

For TC ships on two trips, hire cost runs continuously from trip 1 departure all the way through trip 2 completion. Ensuring that the trip back to Jubail also gets accounted into the costs.

Step 11: Improvement Pass

Check whether any penalised order would be cheaper on a different voyage. For the unavoidably late order, its better to group them on one voyage, as we then reduce fuel costs from 2 voyages.

4.3 Local Exchange

A local exchange is an improvement heuristic which repeatedly makes small local changes to a solution to find a better one. Local exchange allows the algorithm to correct inefficient port assignments across voyages.

Out of the variety of local exchanges, we implemented a move exchange because it directly attacks the biggest inefficiencies in our initial plan.

4.3.1 Local Exchange Algorithm

Step 1: Current Voyage Plan

The algorithm receives the results from the previous constrained optimization problem with a list of ports to visit and the amount of cargo going to each port.

Step 2: Look for a better way to distribute to ports

The algorithm now tries to move a single port from one voyage to another while keeping the same type of vessel. It goes through all voyages of the same vessel type and for each combination it checks whether moving a port improves the solution or not.

Step 3: Check Rules

If in case in step 2 the algorithm finds a better solution, then it checks the following 3 conditions:

1. Capacity condition
2. Temperature zoning
3. Draft condition

Step 4: Optimize and Compare Costs

If the move passes the conditions then both voyages are re-planned with their new route, sailing speed and departure dates.

After doing this, it computes the total cost again and compares the new costs with the old costs. If new costs $<$ old costs, then the new route is accepted and implemented.

Step 5: Repeat until nothing improves

Once a new method is accepted, the algorithm does all the steps again and the search starts from the beginning of the voyage list again.

The whole process repeats until no better solution is found.

4.4 Results & conclusions

After implementing all the algorithms, we found the following results:

Vessel	Mode	Speed(knots)	Route	Departure	Penalty($)	Total Cost ($)
MR50	TC	15	Jubail → Jebel Ali	2024-06-01	0	107.822
MR50	TC	15	Jubail → Karachi	2024-06-01	0	285.075
MR50	TC	15	Jubail → Adabiyah → Paradeep	2024-06-01	0	1.625.404
J19	TC	15	Jubail → Cochin	2024-06-01	0	358.814
J19	TC	15	Jubail → Mumbai	2024-06-01	0	261.362
MR50	TC	15	Jubail → Antwerp	2024-06-01	0	1.947.603
MR50	TC	15	Jubail → Gebze → Ravenna → Rotterdam	2024-06-01	0	2.336.271
J19	TC	15	Jubail → Barcelona	2024-06-01	0	860.611
J19	Spot	15	Jubail → Izmir	2024-06-01	0	371.683
MR50	Spot	15	Jubail → Singapore → HIROHATA	2024-06-01	0	1.120.996
MR50	Spot	15	Jubail → Bangkok → Ulsan	2024-06-01	0	1.251.694
MR50	Spot	15	Jubail → Zhuhai → Jiangyin	2024-06-01	0	1.047.624
MR50	Spot	15	Jubail → Penang → Ningbo	2024-06-01	0	1.003.135
					Total Cost ($)	**12.578.094**

Table 17: Unconstrained Optimization Results

Vessel	Mode	Speed(knots)	Route	Departure	Penalty($)	Total Cost ($)
MR50	TC	15	Jubail → Jebel Ali	2024-06-12	75.000	182.822
MR50	TC	15	Jubail → Karachi	2024-06-19	130.000	415.075
MR50	TC	15	Jubail → Adabiyah → Paradeep	2024-06-06	65.000	1.690.404
J19	TC	15	Jubail → Mumbai	2024-06-09	30.000	291.362
MR50	TC	15	Jubail → Antwerp	2024-06-01	220.000	2.167.603
MR50	TC	15	Jubail → Gebze → Ravenna → Rotterdam	2024-06-01	100.000	2.436.271
J19	TC	15	Jubail → Barcelona	2024-06-01	35.000	895.611
MR50	TC Trip 2	15	Jubail → Singapore → HIROHATA	2024-06-04	10.000	2.098.030
MR50	TC Trip 2	15	Jubail → Bangkok → Ulsan	2024-06-01	115.000	2.313.376
MR50	Spot	15	Jubail → Jiangyin → Zhuhai	2024-06-01	200.000	1.394.076
MR50	Spot	15	Jubail → Penang → Ningbo	2024-06-01	135.000	1.138.135
J19	TC Trip 2	15	Jubail → Cochin	2024-06-02	0	358.814
J19	Spot	15	Jubail → Izmir	2024-06-07	30.000	401.683
					Total Cost ($)	**15.783.262**

Table 18: Constrained Optimization Results

Vessel	Mode	Speed(knots)	Route	Departure	Penalty($)	Total Cost ($)
MR50	TC	15	Jubail → Jebel Ali	2024-06-12	75.000	182.822
MR50	TC	15	Jubail → Karachi	2024-06-19	130.000	415.075
MR50	TC	15	Jubail → Adabiyah	2024-06-05	60.000	976.487
J19	TC	15	Jubail → Mumbai	2024-06-09	30.000	291.362
MR50	TC	15	Jubail → Antwerp	2024-06-01	220.000	2.167.603
MR50	TC	15	Jubail → Gebze → Ravenna → Rotterdam	2024-06-01	100.000	2.436.271
J19	TC	15	Jubail → Barcelona	2024-06-01	35.000	895.611
MR50	TC Trip 2	15	Jubail → Singapore → HIROHATA	2024-06-04	10.000	2.098.030
MR50	TC Trip 2	15	Jubail → Bangkok → Ulsan	2024-06-01	115.000	2.313.376
MR50	Spot	15	Jubail → Jiangyin → Zhuhai	2024-06-01	200.000	1.394.076
MR50	Spot	15	Jubail → Penang → Ningbo → Paradeep	2024-06-01	160.000	1.823.479
J19	TC Trip 2	15	Jubail → Cochin	2024-06-02	0	358.814
J19	Spot	15	Jubail → Izmir	2024-06-07	30.000	401.683
					Total Cost ($)	**15.754.690**

Table 19: Local Exchange Results

4.5 Conclusion

In conclusion, the unconstrained vessel scheduling problem has the lowest cost by far. However, this is a feature of the problem since removing hold constraints and draft limits allow the vessels to go anywhere and transport everything without any issue. This makes the model unrealistic and only serves as a benchmark for the constrained vessel scheduling problem.

On the other hand, after implementing local exchange, we only see a decrease of 0.2% in total costs but this can be explained by our initial constrained vessel scheduling algorithm being close to the local optimum and thus providing little room for improvement for the local exchange.

With little improvement from the local exchange solution, this suggests that improvement can be made to the solution; however this requires us to implement more complicated heuristic models like insertion move, where instead of swapping two vessel assignments, we take one vessel and move it to a different position entirely without matching it with another vessel to trade with.

A logical next step for the entire problem as a whole is to use prices to generate revenues, and to further maximize profit. Price forecasting can also be included to support future planning. Alongside that if possible we should also try and implement insertion move to check whether the 0.2% decrease can be improved further.